Frequency Domain Interpretation of Sampling

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Key Concepts

1) The Fourier transform (FT) of a sampled signal is expressed in terms of the FT of the original continuous-time signal using the following steps:

    a) Define the FT of $x[n]$ to be $X_s(\Omega)$.

        i) Discrete-time frequency $\omega = \Omega T$ where T is the sampling interval.

        ii) Define

$$X_s(\Omega) = X(e^{j\omega})\Big|_{\omega=\Omega T}$$

where $X(e^{j\omega})$ is the discrete-time Fourier transform of $x[n]$. That is,

$$X_s(\Omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega Tn}$$

b) Take the inverse FT of $X_s(\Omega)$ to obtain

$$x_s(t) = \sum_{n=-\infty}^{\infty} x[n]\delta(t - nT)$$

as the continuous-time representation of the sampled signal $x[n]$.

c) Express $X_s(\Omega)$ as a function of $X(\Omega)$, the FT of $x(t)$.

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i) Write $x_s(t) = x(t)s(t)$ where

$$s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

ii) Use multiplication-convolution property of the FT to obtain

$$X_s(\Omega) = \frac{1}{2\pi} X(\Omega) * S(\Omega)$$

iii) Substitute

$$S(\Omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\Omega - k\frac{2\pi}{T})$$

to obtain

$$X_s(\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(\Omega - k\Omega_s)$$

where $\Omega_s = \frac{2\pi}{T}$.

Frequency Domain (FT) Interpretation of Sampling

1) Relate Ω to ω

2) Relate FT of sampled signal to FT of cont. time signal

a) Find FT of a discrete-time signal

$$x[n] \xleftrightarrow{FT} X_s(\Omega)$$

b) Find cont. time rep. for $x[n]$

$$x_s(t) \xleftrightarrow{FT} X_s(\Omega)$$

c) $x_s(t) = f_1(x(t)) \xleftrightarrow{FT} X_s(\Omega) = f_2(X(\Omega))$

1) Relate Ω to ω

Let $g(t) = A\cos(\Omega t + \phi)$, $g[n] = A\cos(\omega n + \phi)$

$g[n] = g(nT) = A\cos(\Omega T n + \phi)$

$$\boxed{\omega = \Omega T}$$

DT freq = CT freq. Sampling interval

rads = $\frac{\text{rads}}{\text{sec}}$. sec

2a) FT of $x[n]$

$$x[n] \xleftrightarrow{\text{DTFT}} X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$\omega = \Omega T$$

$$x[n] \xleftrightarrow{\text{FT}} X_s(\Omega) = X(e^{j\omega})\Big|_{\omega=\Omega T}$$

$$= \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega T n}$$

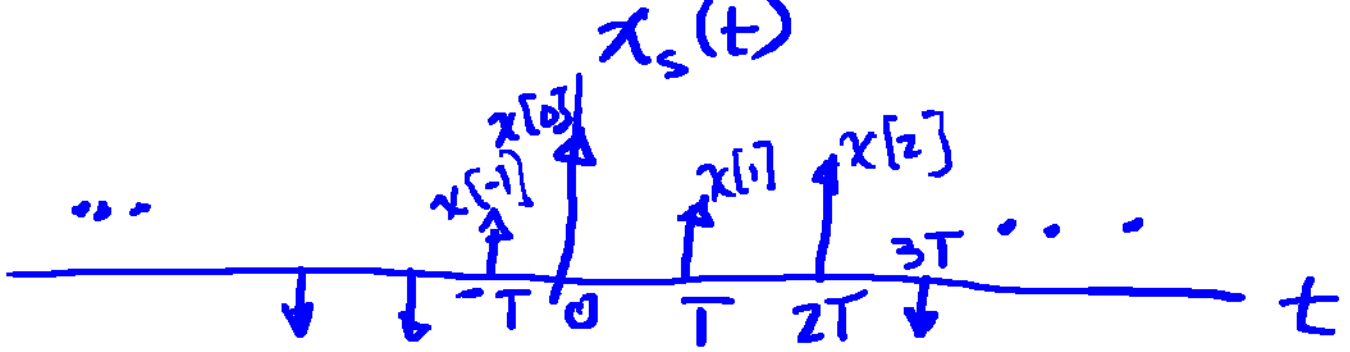
2b) Cont.-time Representation for $x[n]$

$$X_s(\Omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega T n}$$, Find $x_s(t)$

$\delta(t-nT) \xleftrightarrow{FT} e^{-j\Omega T n}$

$$x_s(t) = \sum_{n=-\infty}^{\infty} x[n]\, \delta(t-nT)$$

$x_s(t)$ ↑ CT rep for $x[n]$



2c) Express $X_s(\Omega)$ in terms of $X(\Omega)$

$$x_s(t) = \sum_{n=-\infty}^{\infty} x[n]\,\delta(t-nT)$$

use $x[n] = x(nT)$

$$x(t)\,\delta(t-nT) = x(nT)\,\delta(t-nT)$$

$$= \sum_{n=-\infty}^{\infty} x(nT)\,\delta(t-nT)$$

$$= \sum_{n=-\infty}^{\infty} x(t)\,\delta(t-nT)$$

$$= x(t)\,s(t)$$

where $s(t) = \sum_{n=-\infty}^{\infty} \delta(t-nT)$

$\updownarrow$ FT

$\updownarrow$ FT

$$S(\Omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta\left(\Omega - k\frac{2\pi}{T}\right)$$

$$X_s(\Omega) = \frac{1}{2\pi} X(\Omega) * S(\Omega)$$

$$= \frac{1}{T} \sum_{k=-\infty}^{\infty} X(\Omega) * \delta\left(\Omega - k\frac{2\pi}{T}\right)$$

$$= \frac{1}{T} \sum_{k=-\infty}^{\infty} X\left(\Omega - k\frac{2\pi}{T}\right)$$

Summary

1) $\boxed{\omega = \Omega T}$

2) if $x(t) \xleftrightarrow{FT} X(\Omega)$ and $x[n] = x(nT)$, then

$$x[n] \xleftrightarrow{FT} X_s(\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X\left(\Omega - k\frac{2\pi}{T}\right)$$

